CO2xNxPhrag: Porewater SO4/Cl

May 2026 Samples

2026-07-02

Contents

##Add Required Packages

0.1 Run Information

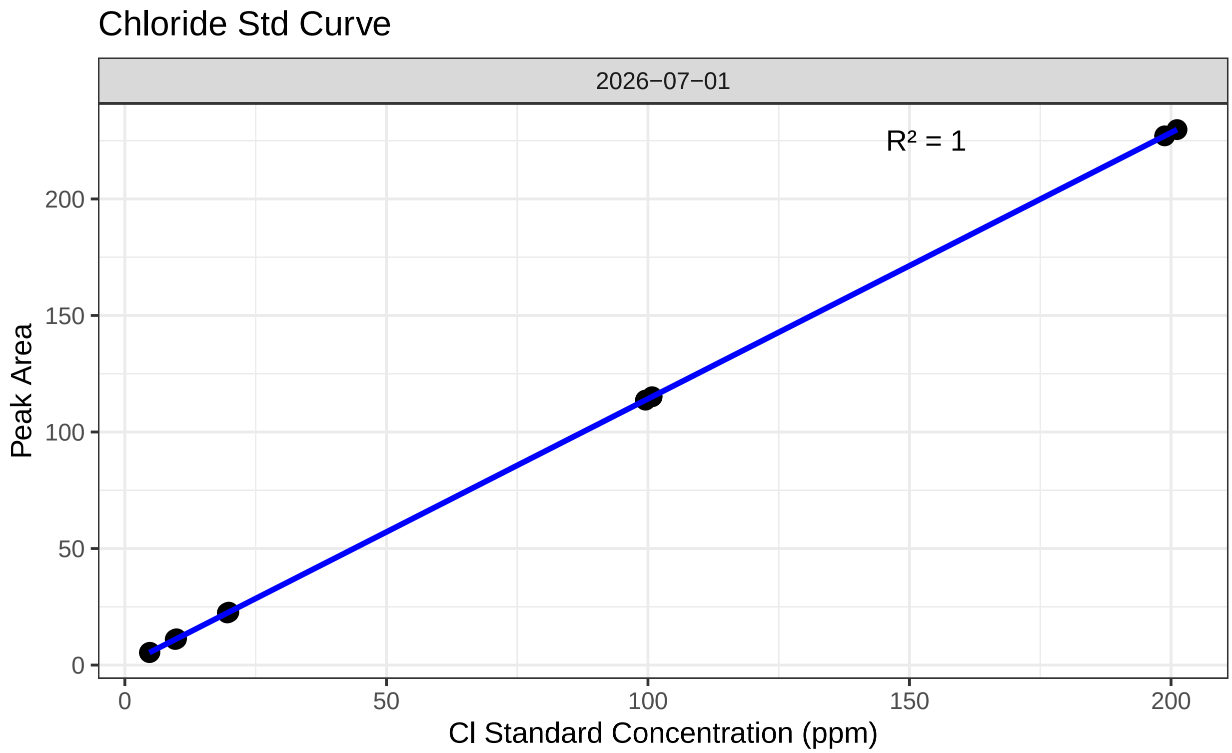
```r
###### Run information - PLEASE CHANGE
  Date_Run = "2026-07-01"  #Date that instrument was run
  Run_by = "Zoe Read & Victoria Ellis"  #Instrument user
  Script_run_by = "Zoe Read & Victoria Ellis" #Code user
  sample_year <- "2026"
  sample_month <- "May"
  Site = "GCReW"
  Project_name = "CO2xNxPhrag"
  run_notes = "
  " #any notes from the run

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/CO2xNxPhrag_202605_Cl.txt"
  raw_file_name_so4 = "Raw Data/CO2xNxPhrag_202605_SO4.txt"

#file path and name of processed data file
  processed_file_name = "Processed Data/GCReW_CO2xNxPhrag_PW_Processed_Cl_SO4_202605.csv"

###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = here(".." , "GCReW", "GCReW_Project_Treatment_Metadata.csv")

  #qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2026.csv"
```

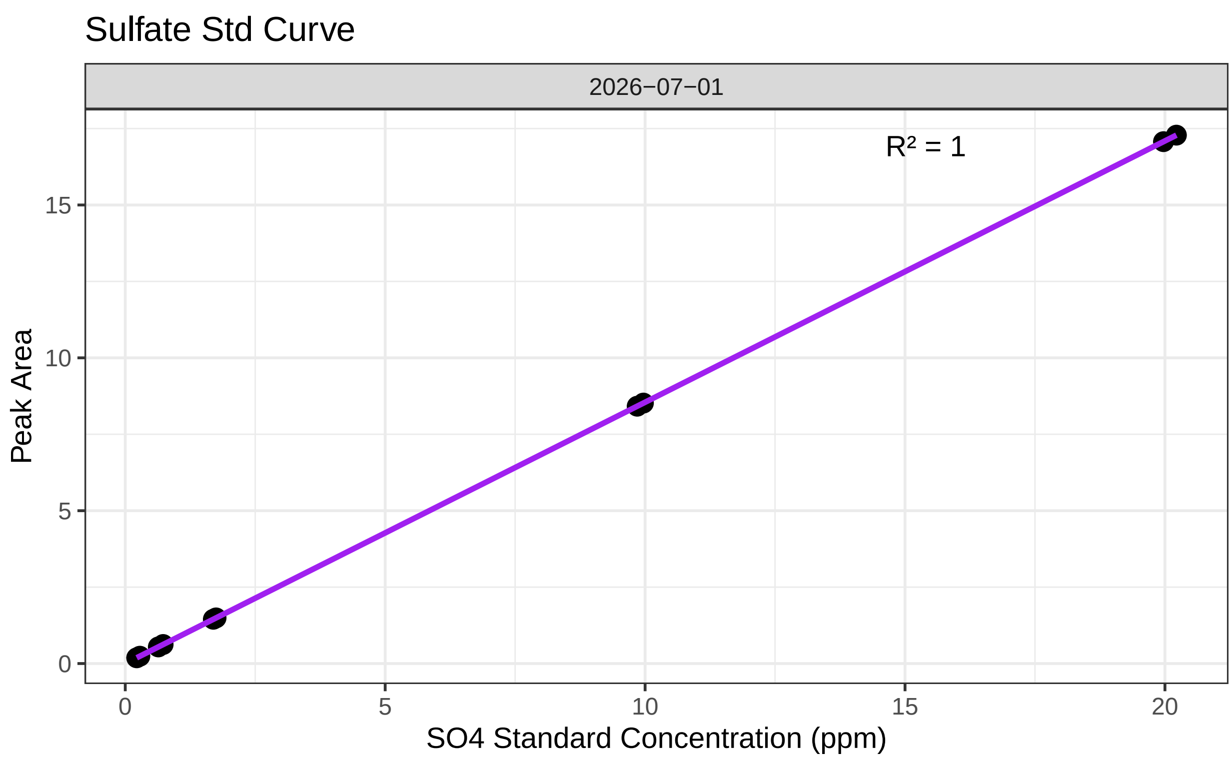
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

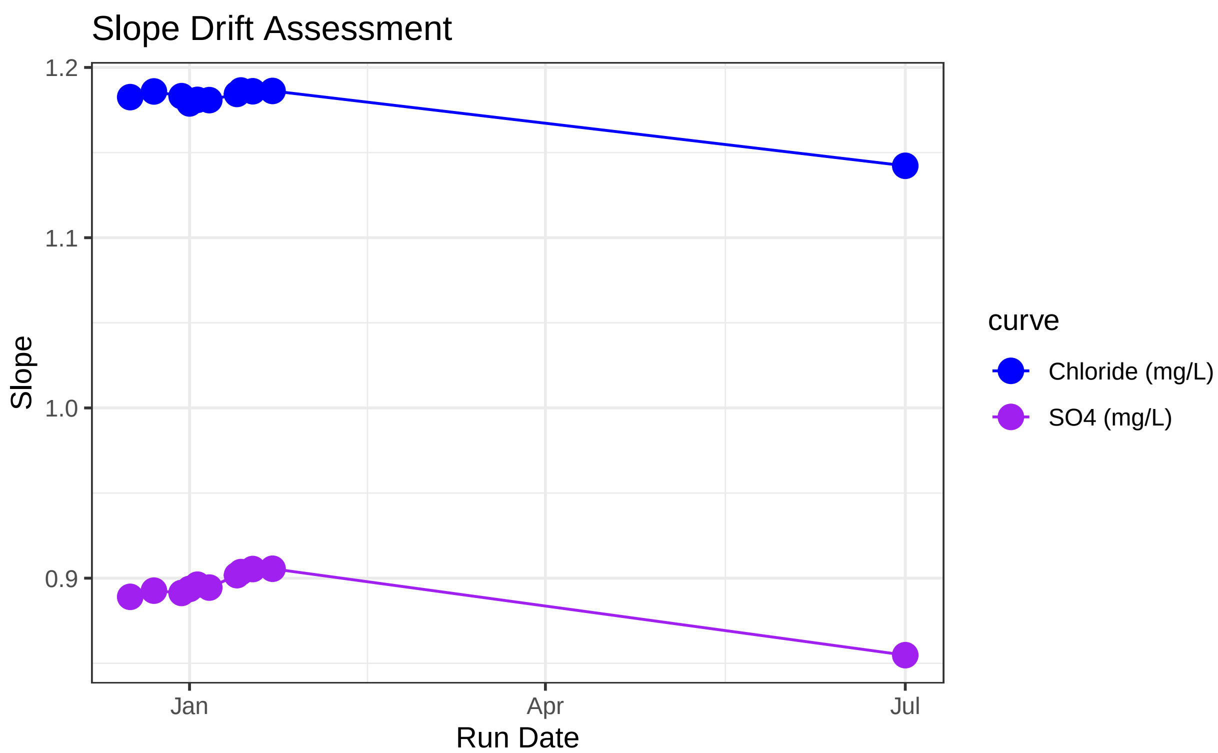
##Import Sample Data

0.2 Assess Standard Curves





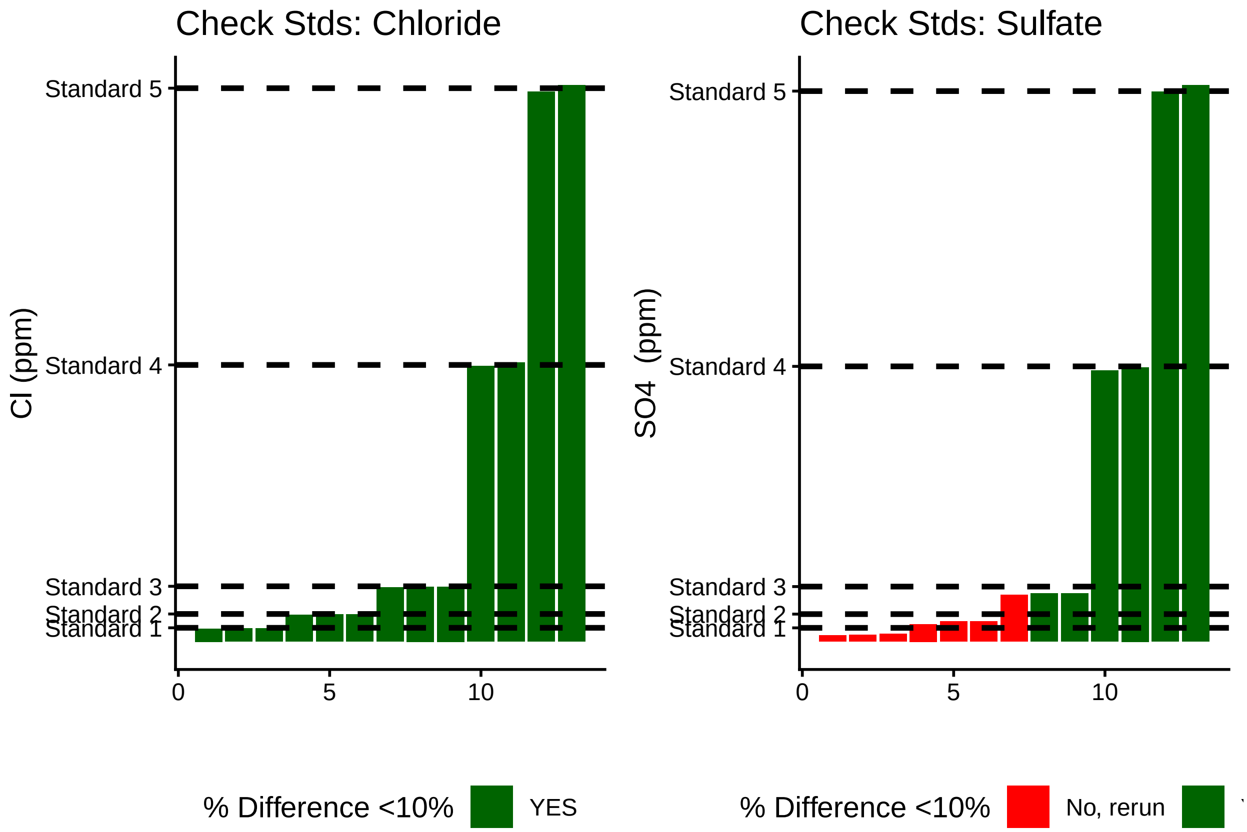

```
## [1] "QAQC log file exists and has been read into the code."
```




```
## [1] "Cl Curve r2 GOOD"

## [1] "SO4 Curve r2 GOOD"
```

0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl  sd_Cl   cv_Cl flag_Cl
##   <chr>        <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    4.74 0.0437 0.00921 Chloride Check Standard RSD within Range - ~
## 2 Standard 2    9.77 0.130  0.0133  Chloride Check Standard RSD within Range - ~
## 3 Standard 3   19.8  0.166  0.00840 Chloride Check Standard RSD within Range - ~
## 4 Standard 4  100.   0.901  0.00899 Chloride Check Standard RSD within Range - ~
## 5 Standard 5  200.   1.66   0.00832 Chloride Check Standard RSD within Range - ~

## # A tibble: 5 x 5
##   sample_ID  mean_SO4 sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    0.250 0.0302 0.121   Sulfate Check Standard RSD within Range - ~
## 2 Standard 2    0.700 0.0552 0.0789  Sulfate Check Standard RSD within Range - ~
## 3 Standard 3    1.73  0.0345 0.0199  Sulfate Check Standard RSD within Range - ~
## 4 Standard 4    9.91  0.0842 0.00850 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5   20.1   0.176  0.00875 Sulfate Check Standard RSD within Range - ~

## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"

## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



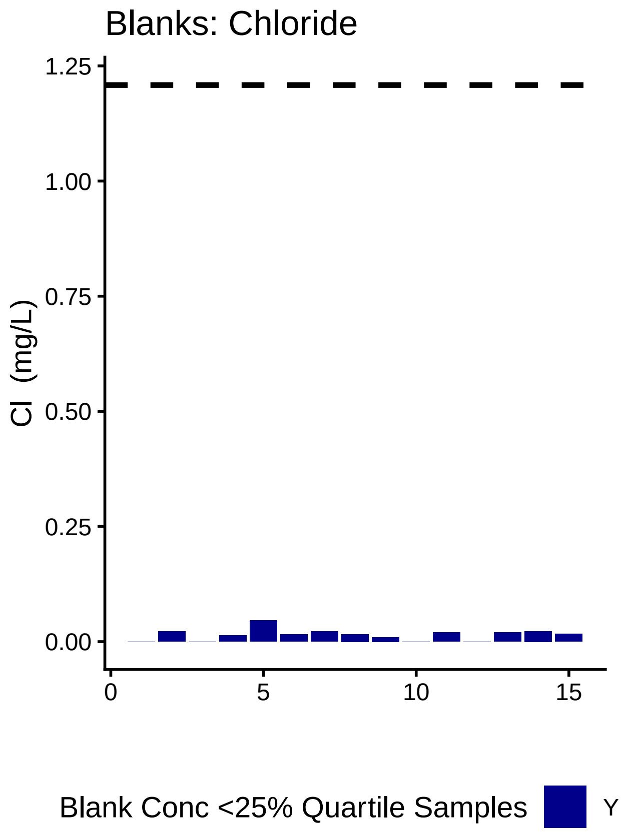

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

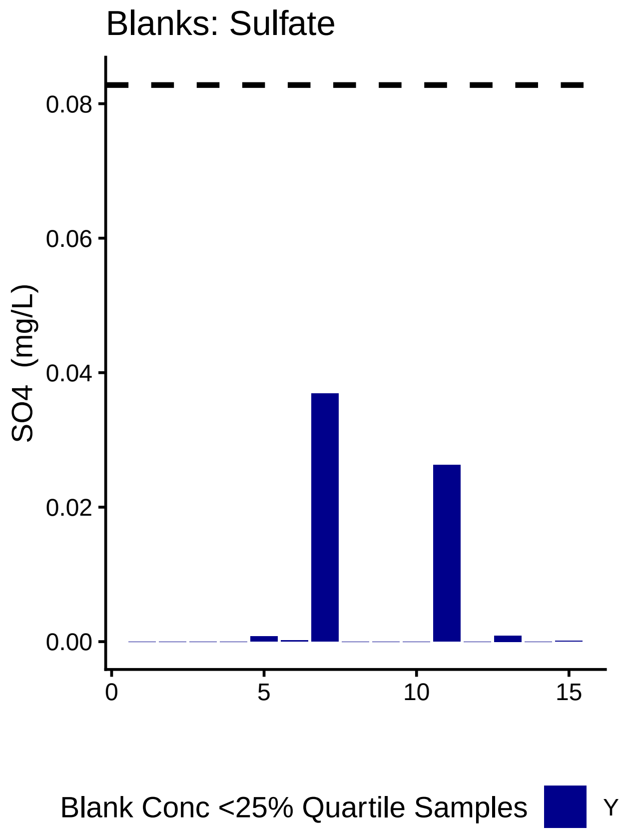
```
## [1] "<80% of Sulfate Check Standards are within range of expected concentration - REASSESS"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

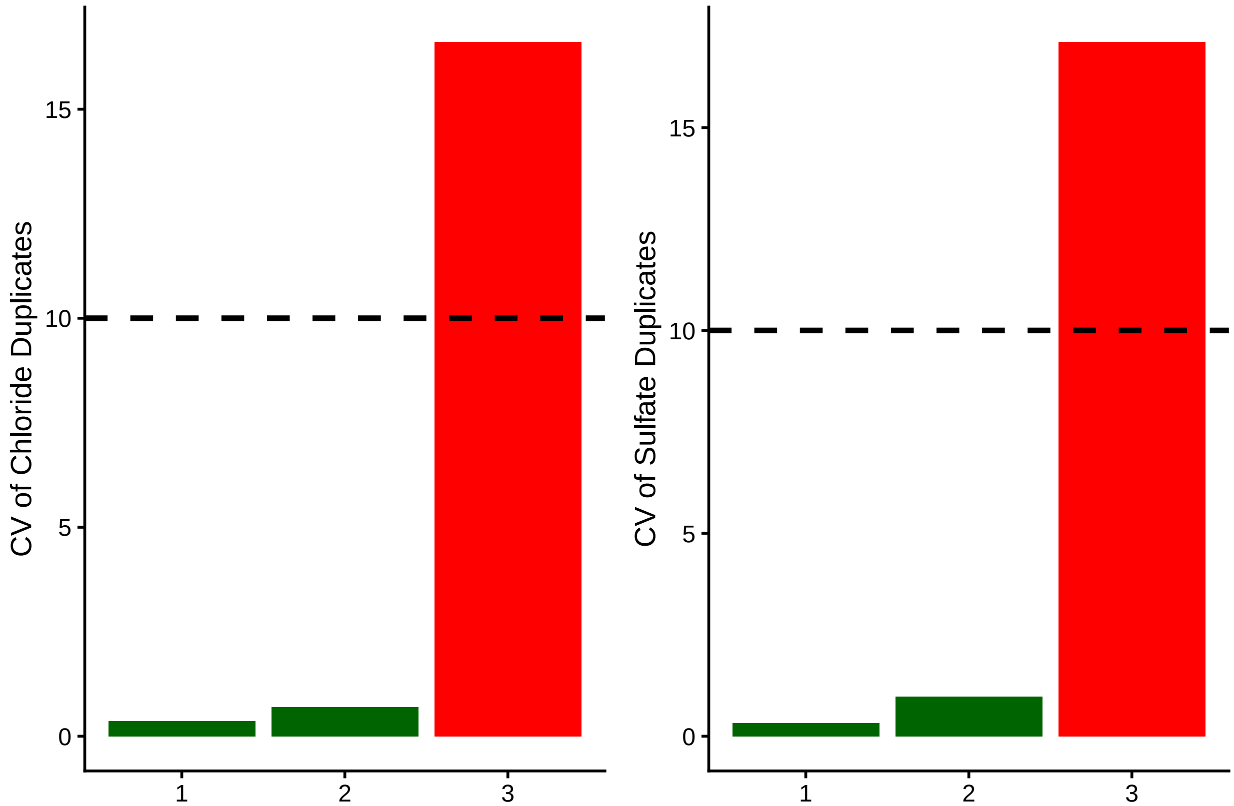
```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```





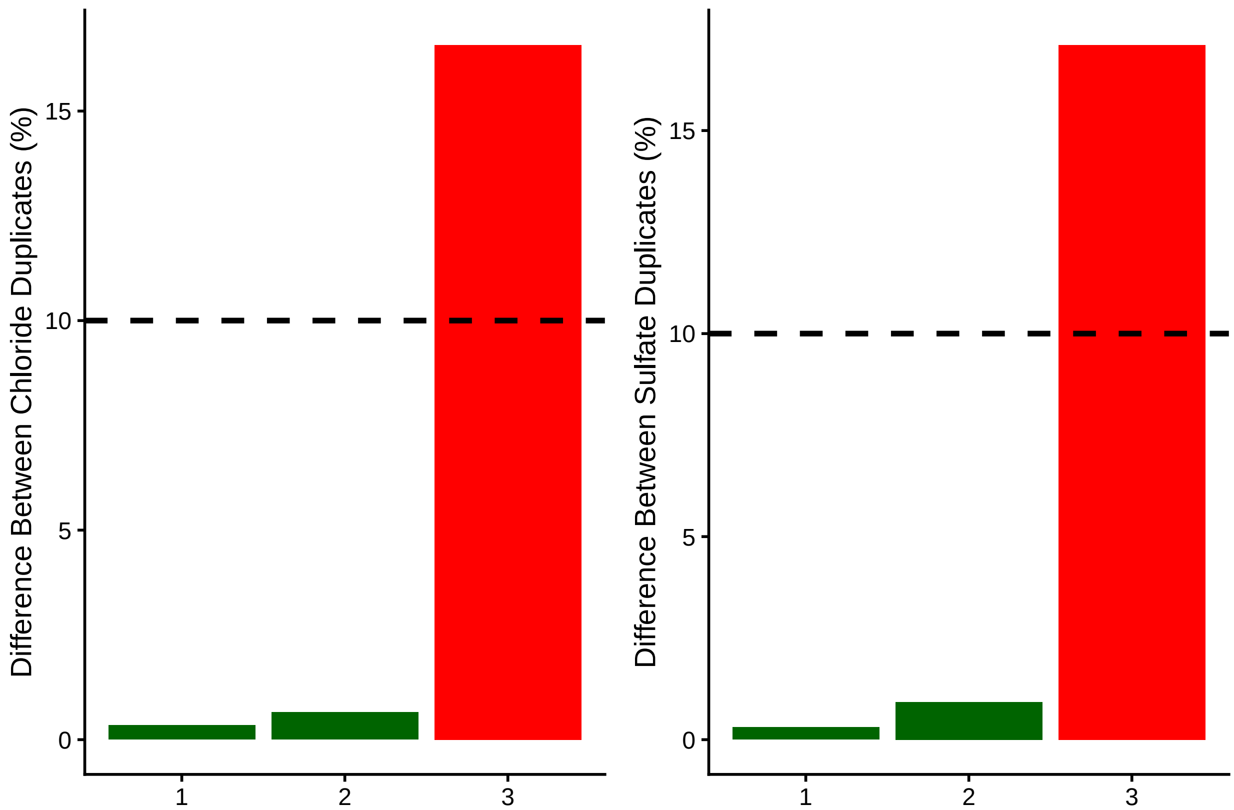

```
## Chloride blanks mean ppm:

## [1] 0.01504667

## Sulfate blanks mean ppm:

## [1] 0.004346667
```

0.5 Assess Duplicates




```
## [1] "<80% of Chloride Duplicates have a CV <10% - REASSESS"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

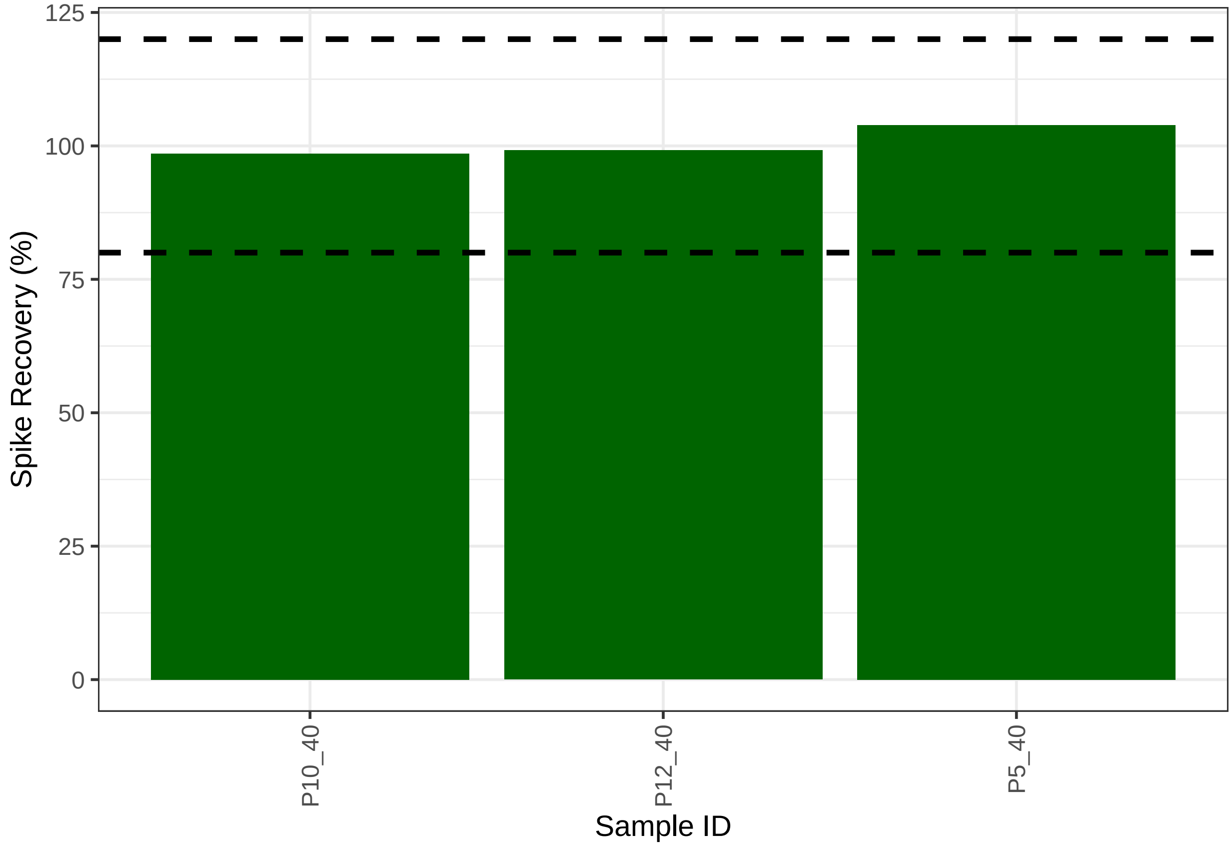



```
## [1] "<80% of Chloride Duplicates have a CV <10% - REASSESS"

## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

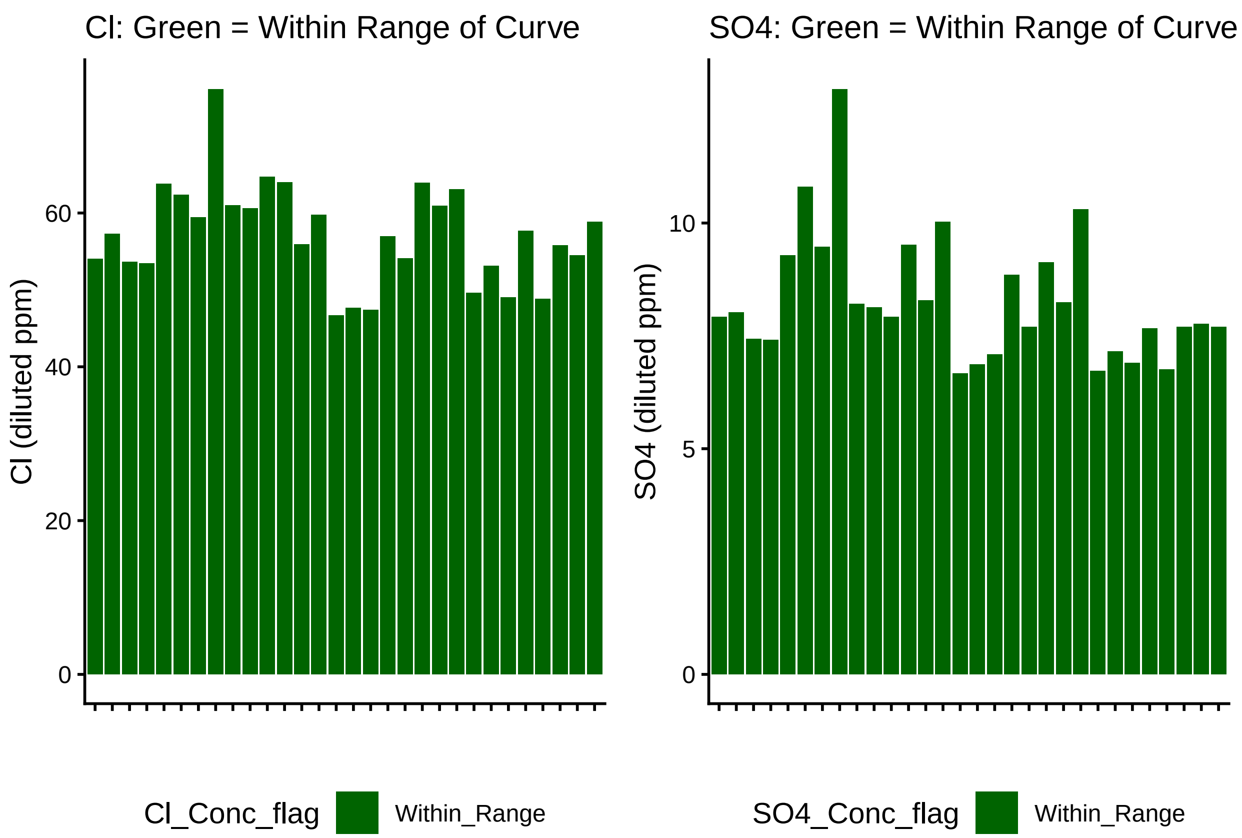


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

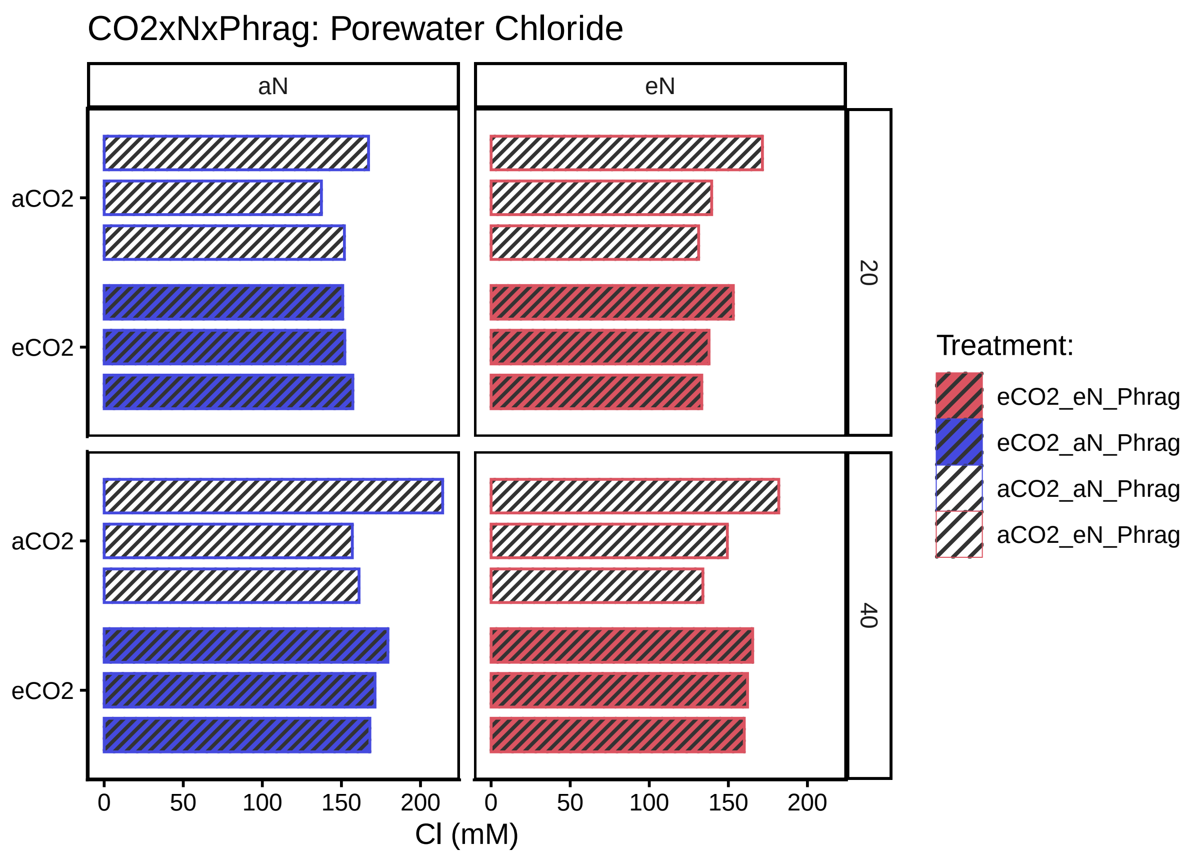
Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Check to see if samples run match metadata & merge info

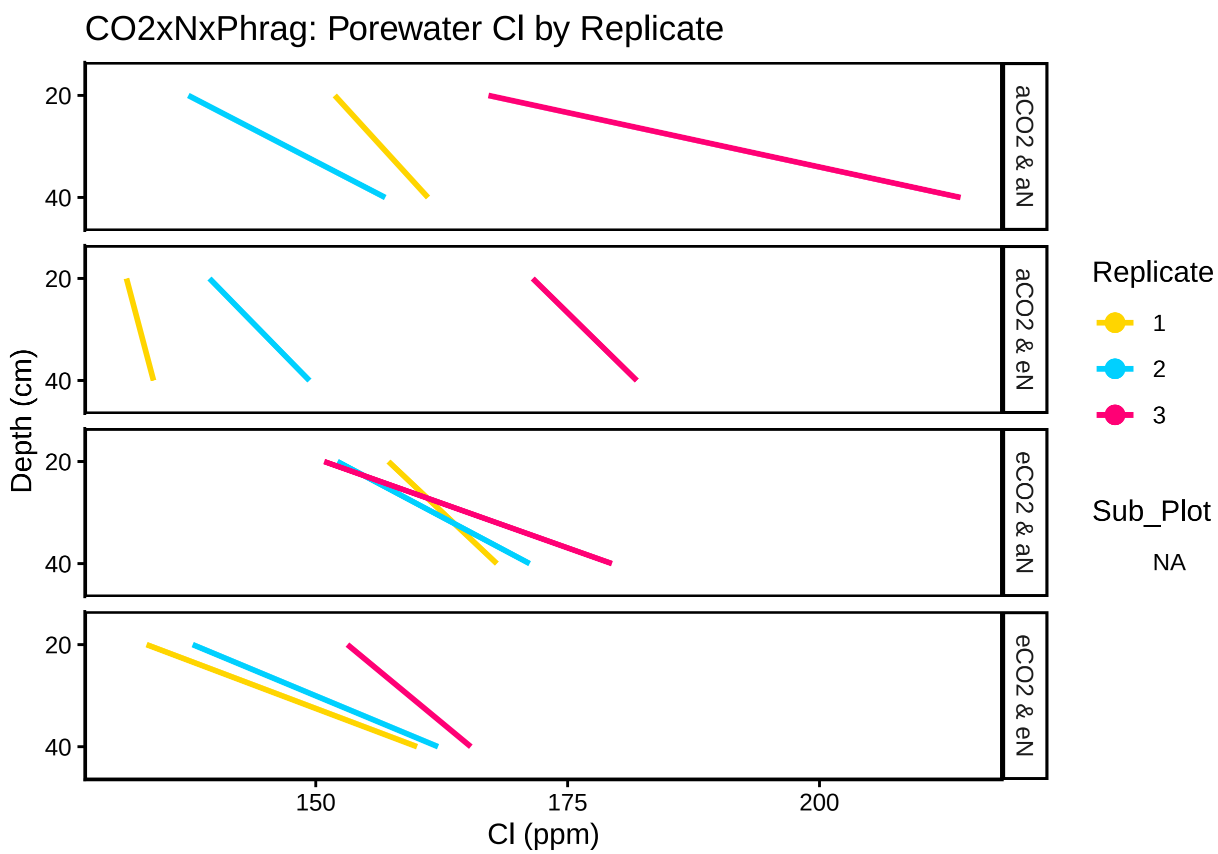
```
## All sample IDs are present in metadata.
```

0.10 Visualize Data

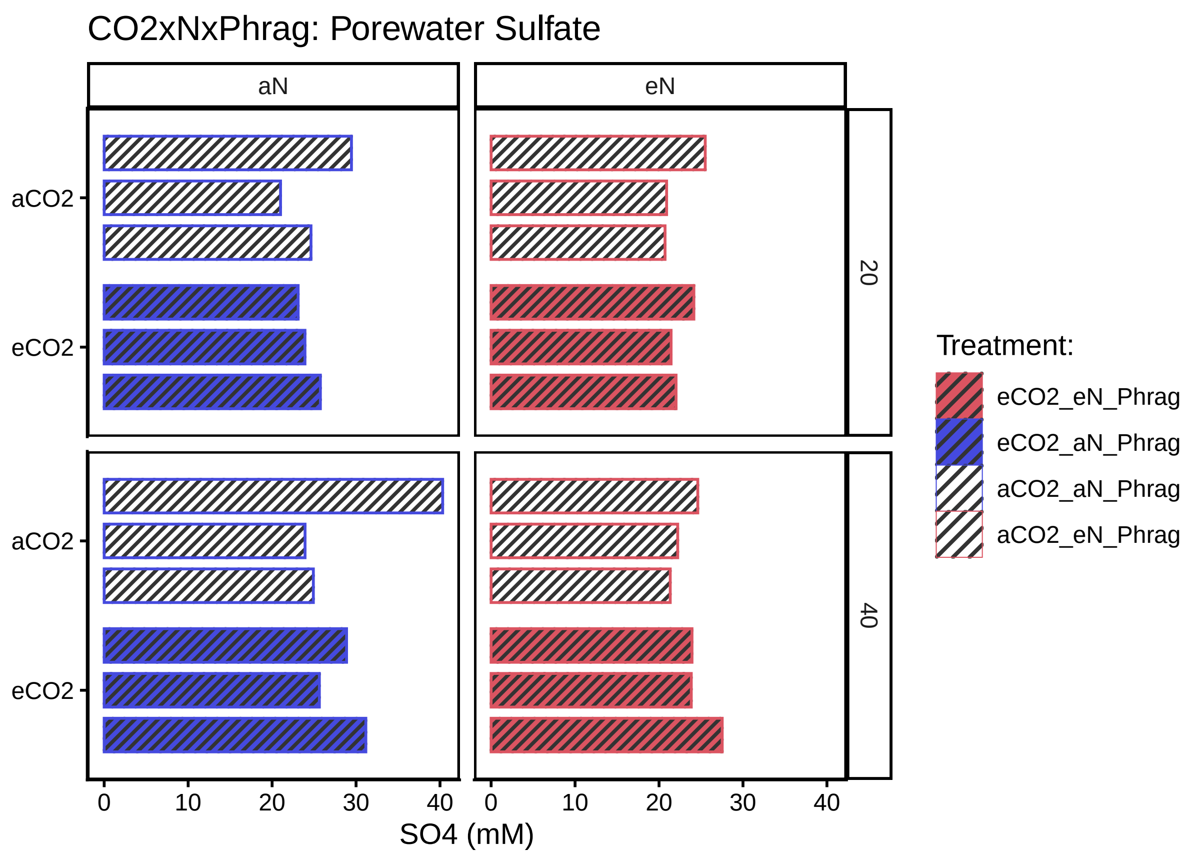
```
## Visualize Data
```



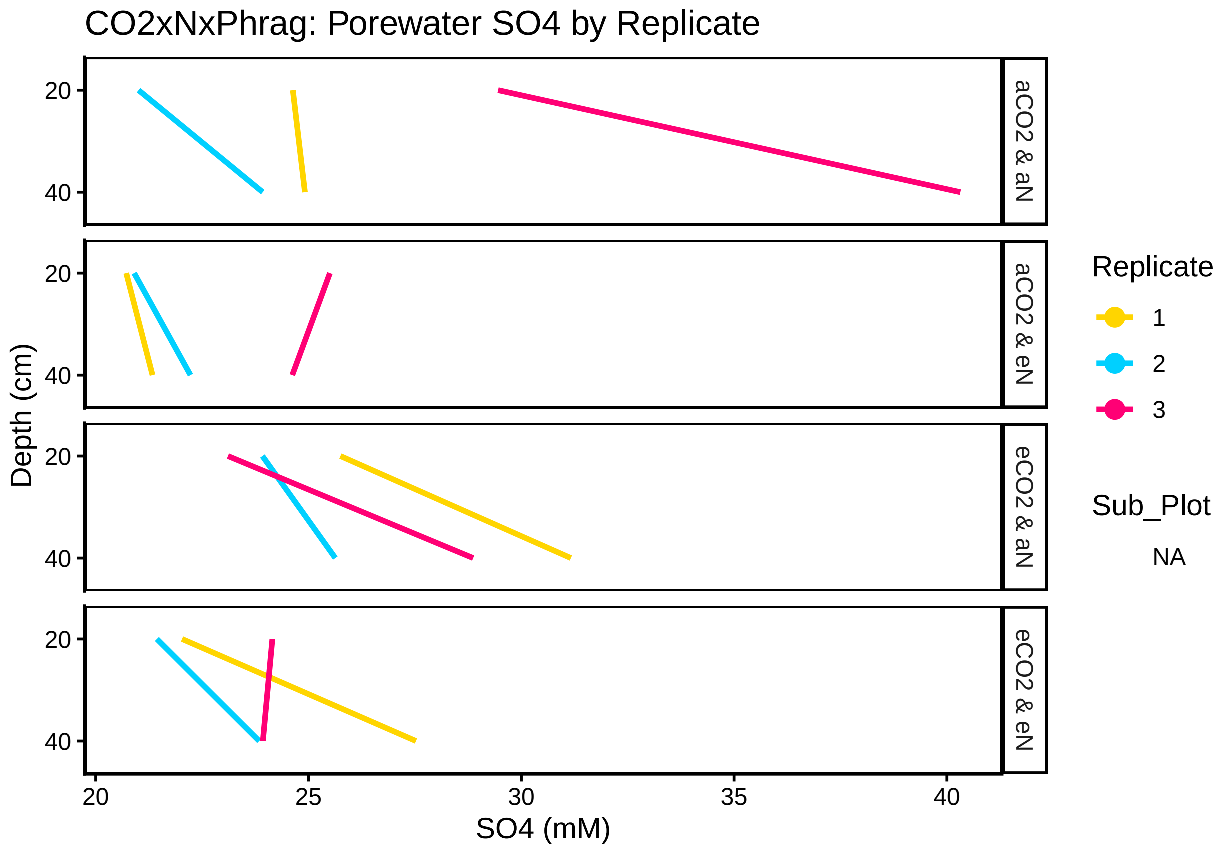

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Visualize SO4 Data




```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



0.11 Export Processed Data

#end